Probability of Single Events

Name _____ Date _____ No calculator Version 1

KEY WORDS probability · outcome · event · equally likely · certain · impossible · expected number

A The probability scale ●○○

ANSWER ON THE SHEET

Remember: A probability is a number from 0 (impossible) to 1 (certain). An even chance is $\frac{1}{2}$.

1. Choose the best description of each event: impossible, unlikely, even chance, likely or certain.

a) Rolling a 7 on an ordinary dice _____

b) Getting a head when a coin is tossed _____

c) Rolling a number less than 7 on a dice _____

d) Rolling a 6 on a dice _____

e) Picking a red counter from a bag of 9 red and 1 blue _____

f) Rolling an even number on a dice _____

B Equally likely outcomes ●○○

WORK IN YOUR BOOK

Remember: Probability of an event = the number of ways it can happen ÷ the total number of outcomes. Give fractions in their simplest form.

2. An ordinary dice is rolled. Find the probability of rolling:

a) a 5 **b)** an even number **c)** a number greater than 4

d) a prime number **e)** a number less than 7 **f)** an 8

3. A bag holds 5 red, 3 blue and 2 green counters. One counter is taken out without looking. Find the probability that it is:

a) red **b)** blue **c)** not green **d)** yellow

4. Cards numbered 1 to 20 are shuffled and one card is picked. Find the probability that the number is:

a) a multiple of 5 **b)** a square number **c)** a prime number **d)** greater than 15

C The probability of not happening ●●○

WORK IN YOUR BOOK

Remember: An event either happens or it does not, so $P(\text{not } A) = 1 - P(A)$. With percentages, subtract from 100%.

5. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.3 **b)** $\frac{2}{5}$ **c)** 35%

d) 0.82 **e)** $\frac{7}{12}$ **f)** 4%

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D One event or another ●●○ WORK IN YOUR BOOK

Remember: If two events cannot both happen at the same time, add their probabilities: $P(A \text{ or } B) = P(A) + P(B)$.

6. Find the probability of getting:

a) a 2 or a 5 on a dice

b) a 1 or an even number on a dice

c) a red or a green counter from a bag holding 5 red, 3 blue and 2 green counters

d) a number less than 3 or greater than 15 from cards numbered 1 to 20

E Expected number ●●○ WORK IN YOUR BOOK

Remember: Expected number = probability × number of trials.

7. Work out how many times you would expect each event to happen.

a) A fair dice is rolled 120 times. How many sixes would you expect?

b) A coin is tossed 250 times. How many heads would you expect?

c) The probability that a spinner lands on red is 0.35. It is spun 200 times. How many reds would you expect?

d) The probability that a bus is late is 0.08. How many late buses would you expect in 50 days?

e) A bag holds 3 blue and 5 white counters. A counter is picked and put back 400 times. How many blues would you expect?

f) The probability of winning a prize is $\frac{1}{20}$. How many prizes would you expect from 300 tickets?

F Problems ●●● WORK IN YOUR BOOK

8. A box holds 24 chocolates. The probability of picking a dark chocolate is $\frac{3}{8}$. How many dark chocolates are in the box?

9. Three runners are in a race. The probability that Ana wins is 0.25 and the probability that Ben wins is 0.4. Find the probability that the third runner, Caleb, wins.

10. A bag holds 4 red counters, 6 blue counters and some green counters. The probability of picking a green counter is $\frac{1}{3}$. How many green counters are there?

11. A spinner has 5 equal sections, and 2 of them are red. It is spun 80 times. How many times would you expect it not to land on red?

★ Extension FOR EARLY FINISHERS

12. Bag A holds 8 counters and 3 of them are red. Bag B holds 12 counters and 5 of them are red. Which bag gives the better chance of picking a red counter? Show your working.

13. A dice is rolled. Find the probability that the number is prime or even (or both).

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Probability of Single Events

Name _____ Date _____ No calculator Version 2

KEY WORDS probability · outcome · event · equally likely · certain · impossible · expected number

A The probability scale ●○○ — ANSWER ON THE SHEET

Remember: A probability is a number from 0 (impossible) to 1 (certain). An even chance is $\frac{1}{2}$.

1. Choose the best description of each event: impossible, unlikely, even chance, likely or certain.

a) Rolling a 0 on an ordinary dice _____

b) Getting a tail when a coin is tossed _____

c) Rolling a number greater than 0 on a dice _____

d) Rolling a 1 on a dice _____

e) Picking a blue counter from a bag of 8 blue and 2 red _____

f) Rolling an odd number on a dice _____

B Equally likely outcomes ●○○ — WORK IN YOUR BOOK

Remember: Probability of an event = the number of ways it can happen ÷ the total number of outcomes. Give fractions in their simplest form.

2. An ordinary dice is rolled. Find the probability of rolling:

a) a 2 **b)** an odd number **c)** a number less than 3

d) a multiple of 3 **e)** a number greater than 0 **f)** a 9

3. A bag holds 4 red, 6 blue and 2 green counters. One counter is taken out without looking. Find the probability that it is:

a) red **b)** blue **c)** not green **d)** yellow

4. Cards numbered 1 to 24 are shuffled and one card is picked. Find the probability that the number is:

a) a multiple of 5 **b)** a square number **c)** a prime number **d)** greater than 15

C The probability of not happening ●●○ — WORK IN YOUR BOOK

Remember: An event either happens or it does not, so $P(\text{not } A) = 1 - P(A)$. With percentages, subtract from 100%.

5. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.4 **b)** $\frac{3}{7}$ **c)** 45%

d) 0.76 **e)** $\frac{5}{12}$ **f)** 8%

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D One event or another ●●○ WORK IN YOUR BOOK

> **Remember:** If two events cannot both happen at the same time, add their probabilities: $P(A \text{ or } B) = P(A) + P(B)$.

6. Find the probability of getting:

a) a 3 or a 4 on a dice

b) a 1 or an even number on a dice

c) a red or a green counter from a bag holding 4 red, 6 blue and 2 green counters

d) a number less than 4 or greater than 16 from cards numbered 1 to 20

E Expected number ●●○ WORK IN YOUR BOOK

> **Remember:** Expected number = probability × number of trials.

7. Work out how many times you would expect each event to happen.

a) A fair dice is rolled 180 times. How many fours would you expect?

b) A coin is tossed 350 times. How many tails would you expect?

c) The probability that a spinner lands on blue is 0.45. It is spun 200 times. How many blues would you expect?

d) The probability that a train is late is 0.06. How many late trains would you expect in 50 days?

e) A bag holds 5 blue and 3 white counters. A counter is picked and put back 240 times. How many blues would you expect?

f) The probability of winning a prize is $\frac{1}{25}$. How many prizes would you expect from 400 tickets?

F Problems ●●● WORK IN YOUR BOOK

8. A box holds 30 chocolates. The probability of picking a dark chocolate is $\frac{2}{5}$. How many dark chocolates are in the box?

9. Three runners are in a race. The probability that Ana wins is 0.3 and the probability that Ben wins is 0.45. Find the probability that the third runner, Caleb, wins.

10. A bag holds 5 red counters, 7 blue counters and some green counters. The probability of picking a green counter is $\frac{1}{3}$. How many green counters are there?

11. A spinner has 8 equal sections, and 3 of them are red. It is spun 120 times. How many times would you expect it not to land on red?

★ Extension FOR EARLY FINISHERS

12. Bag A holds 10 counters and 4 of them are red. Bag B holds 14 counters and 5 of them are red. Which bag gives the better chance of picking a red counter? Show your working.

13. A dice is rolled. Find the probability that the number is odd or greater than 4 (or both).

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